

## **Question 1**

### **a. What is a Firewall? and what purpose would it serve?**

A firewall is a security system (hardware, software, or both) that monitors and controls traffic entering and leaving a computer or network. It keeps trusted internal networks separate from untrusted external networks, like the Internet. The main goal is to prevent unauthorized access to systems and protect them from cyber threats.

A firewall looks at network traffic and uses a set of rules to protect the network. Traffic entering or leaving is allowed to pass through if it follows the rules; otherwise, it is stopped. For instance, a firewall can block connections from unknown sources that seem suspicious or prevent certain apps from accessing the internet.

The main purposes of a firewall are

- Monitoring network traffic to detect suspicious or malicious activity
- Blocking unauthorized access from hackers or malicious software
- Filtering incoming and outgoing data packets based on security rules
- Acting as a first line of defence for computers and networks connected to the internet

Overall, a firewall is an essential part of cybersecurity because it helps protect sensitive information and reduces the risk of cyberattacks.

### **b. What threats a Firewall does NOT protect you against?**

Although firewalls are important security tools, they do not protect against every type of cyber threat. They mainly control network traffic, but many attacks can still occur through other methods.

Some threats that firewalls do not fully protect against are

Malware from email attachments or downloads

- If a user downloads an infected file or clicks a malicious link in an email, the malware can still install itself even if a firewall is in place.

Phishing and social engineering attacks

- Users can still fall victim to phishing and social engineering attacks, which can lead them to disclose passwords or personal information via fake emails or websites, despite the presence of firewalls.

Viruses or infected files already inside the network

- Once malware enters a device, it can spread within the system or network even if a firewall is installed.

Weak passwords or user mistakes

- If someone shares their password or uses a weak one, a firewall cannot prevent them from logging in.

Because of these limitations, firewalls are usually combined with other security tools such as antivirus software, secure passwords, and user awareness training.

## **Question 2**

### **a. What is a VPN (Virtual Private Network)? What purpose does it serve?**

A VPN, or virtual private network, is a technology that creates a secure, encrypted connection between a user's device and the internet. It lets people send data through a secure "tunnel" that prevents others from seeing or reading it.

The main reason to use a VPN is to protect your privacy and safety when you surf the web, especially on unsecured networks. When a VPN is on, the user's internet traffic is encrypted before it goes across the network. This means that hackers, internet service providers, or other people on the same network can't easily see what the user is doing.

Common purposes of using a VPN include:

- Encrypting internet traffic to protect sensitive information
- Providing secure remote access to private networks
- Protecting privacy when using public Wi-Fi networks
- Preventing others from monitoring online activities

Overall, VPNs help increase online privacy and reduce the risk of data being intercepted by cybercriminals.

### **b. When connecting to an Unsecured Wireless Network, like at a mall or a store or airport, what steps should you take to protect your information?**

Public Wi-Fi networks in places such as malls, airports, and cafes are convenient but often not secure. Because many users share the same network, attackers may try to intercept personal data.

To protect your information when using public Wi-Fi, several precautions should be taken

Use a VPN whenever possible.

- A VPN encrypts your internet traffic, making it much harder for attackers on the network to read your data.

Only visit secure websites (HTTPS).

- Secure websites encrypt the data sent between your browser and the website. You can identify them by "https" and a lock icon in the address bar.

Avoid accessing sensitive accounts.

- Do not perform banking transactions or access confidential information while on public Wi-Fi.

Log out of accounts when finished.

- Staying logged in can make it easier for attackers to access your accounts.

Keep your device updated and use security software.

- Updates and antivirus programs help protect your device from known vulnerabilities and malware.

By following these precautions, users can significantly reduce the risks associated with unsecured wireless networks.

## References

### Question 1

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